

Data Card: Agricultural Extension RAG Corpus

1. Dataset Overview

- **Dataset Name:** Agricultural Extension RAG Corpus (TRI AI Cohort 10 Benchmark)
- **Domain:** Agronomy, Crop Protection, Plant Pathology, Soil Health, and Extension Services
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- **Geographic Focus:** Sub-Saharan Africa (Localized pest management, regional crop varieties, and regional chemical/fertilizer formulations)
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- **Total Documents:** 695 structured factsheets and extension passages
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- **Evaluation Queries:** 200 real-world agricultural queries submitted by smallholder extension workers
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2. Source & Provenance

- **Data Providers:** Vetted extension manuals and advisory factsheets compiled from international and regional bodies (including FAO, CGIAR, and Plantwise).
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- **Licensing & Usage:** Publicly available educational and extension resources curated for academic benchmarking and non-commercial research under the TRI AI Cohort 10 Competition.
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3. Schema & Data Fields

Field Name	Type	Description	Handling Strategy
document_id	Integer / String	Unique primary identifier for each passage	Preserved as exact string key
title	String	Factsheet or advisory section heading	Preserved; included in schema enrichment

crop	String	Target crop category (e.g., Maize, Cassava, Rice, General)	Imputed to 'general' if missing
source	String	Publishing authority or advisory body	Preserved; included in schema enrichment
text	String	Core body text containing diagnostic and treatment advice	Cleaned of whitespace while preserving chemical NPK codes

4. Text Enrichment & Preprocessing Pipeline

To maximize cross-encoder and vector attention alignment, all document fields are combined into a standardized schema string using `src/preprocess.py`:

Domain-Specific Guardrails:

- **Preservation of Technical Terminology:** Alphanumeric tokens, pesticide chemical names, and hyphenated NPK fertilizer ratios (e.g., 15-15-15, NPK-20-10-10) are preserved using regex tokenization `(\b\w+(?:-\w+)*\b)`.
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- **Noise Removal:** Unstructured HTML markup, double spaces, and redundant trailing spaces are stripped out.
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5. Evaluation & Task Setup

- **Primary Task:** Passages are retrieved and ranked to maximize Mean Reciprocal Rank (**MRR@10**) and **nDCG@5** for 200 evaluation queries.
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- **Output Standard:** For each query, the top 5 highest-confidence DocumentId matches are exported in order, yielding exactly 1,000 output rows in `submission.csv`.